

7.8b Improper Integrals - continued

Direct Comparison Test

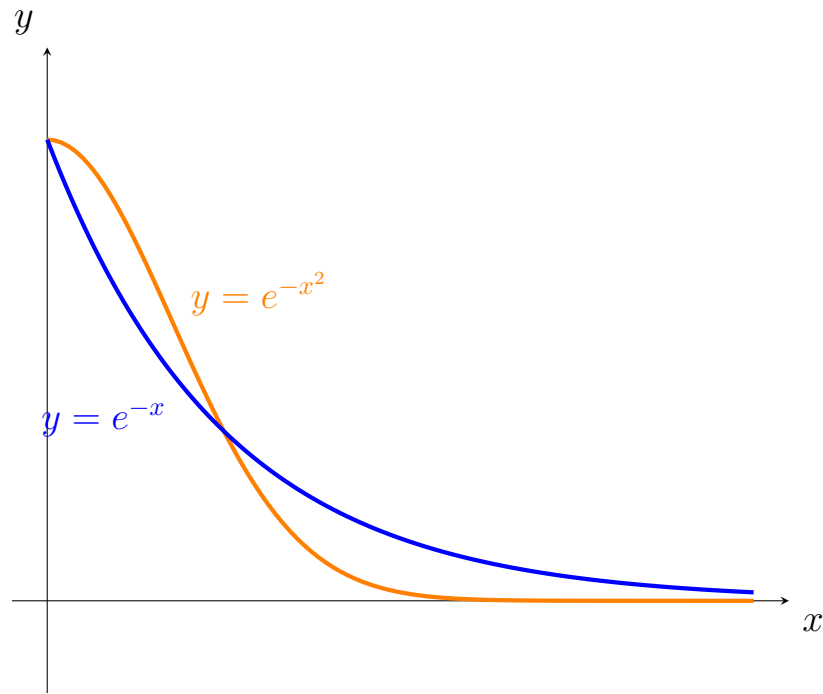
Let f and g be continuous on $[a, \infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \geq a$.
Then

1. $\int_a^\infty f(x) dx$ _____ if $\int_a^\infty g(x) dx$ _____
2. $\int_a^\infty g(x) dx$ _____ if $\int_a^\infty f(x) dx$ _____

Why does this make sense?

Example 6

Does $\int_0^{\infty} e^{-x^2} dx$ converge?



Example 7

Apply the Direct Comparison Test to the following integrals to determine convergence/divergence.

1. $\int_1^{\infty} \frac{\sin^2(x)}{x^2} dx$

2. $\int_{2\pi}^{\infty} \frac{1}{\sqrt{x^2 - \pi^2}} dx$

(Extra) Example 8

Which of the following are improper integrals? Justify your answer.

1. $\int_{-\infty}^3 \frac{1}{x^2 + 2} dx$

2. $\int_0^7 \ln(x) dx$

3. $\int_{-1}^4 \frac{1}{x} dx$

4. $\int_{-5}^2 \sqrt{4 - x^2} dx$

5. $\int_0^{\pi/4} \sec(x) dx$